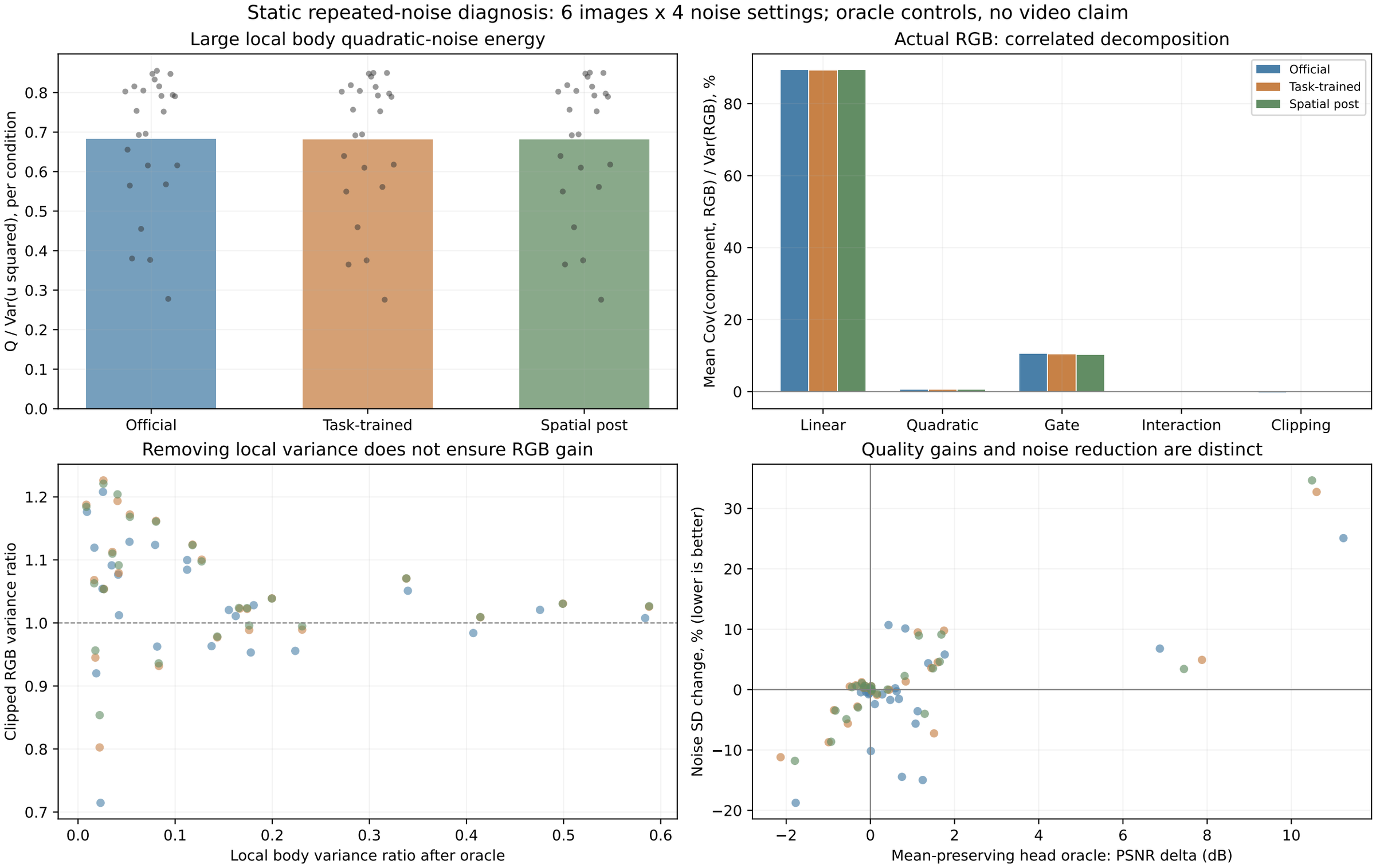
Static repeated-noise diagnosis: 6 images x 4 noise settings; oracle controls, no video claim
Large local body quadratic-noise energy
Q / Var(u squared), per condition
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Official
Task-trained
Spatial post
Actual RGB: correlated decomposition
Mean Cov(component, RGB) / Var(RGB), %
0
20
40
60
80
Official
Task-trained
Spatial post
Linear
Quadratic
Gate
Interaction
Clipping
Removing local variance does not ensure RGB gain
Clipped RGB variance ratio
0.7
0.8
0.9
1.0
1.1
1.2
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Local body variance ratio after oracle
Quality gains and noise reduction are distinct
Noise SD change, % (lower is better)
−20
−10
0
10
20
30
−2
0
2
4
6
8
10
Mean-preserving head oracle: PSNR delta (dB)